

Introduction to Firewall TASK – 7

- 1. Install firewalld on your Linux system, start the service, and verify its status using systemctl and firewall-cmd commands.**

Ans :

Command	Purpose	Result
sudo apt install -y firewalld	Install firewalld	Installed successfully
sudo systemctl start firewalld	Start service	Service started
sudo systemctl status firewalld	Verify service status	Active (running)
sudo firewall-cmd --state	Verify firewall daemon	running
sudo firewall-cmd --get-active-zones	Show active zone	public
sudo firewall-cmd --list-all	Display current rules	Rules displayed successfully

- 2. Add a new rule to firewalld to allow incoming HTTP (port 80) traffic, then use firewall-cmd to confirm the rule is active.

Hint: Use the --permanent and --reload options to make changes persistent.**

Ans :

Command	Purpose	Result
<code>sudo firewall-cmd --permanent --add-service=http</code>	Allow HTTP traffic permanently	success
<code>sudo firewall-cmd --reload</code>	Apply permanent changes	success
<code>sudo firewall-cmd --list-services</code>	Verify allowed services	http listed
<code>sudo firewall-cmd --list-all</code>	Display active firewall configuration	HTTP service visible

3. Temporarily block all incoming traffic on port 8080 using firewalld, then test by attempting to access a simple web server (e.g., Python's http.server) running on that port.

Ans :

Command	Purpose	Result
<code>python3 -m http.server 8080</code>	Start test web server	Server started successfully
<code>curl http://localhost:8080</code>	Verify connectivity before blocking	Page loaded successfully
<code>sudo firewall-cmd --add-rich-rule='rule family="ipv4" port</code>	Block incoming	Rule added successfully

Command	Purpose	Result
port="8080" protocol="tcp" reject'	traffic on port 8080	
sudo firewall-cmd --list-rich-rules	Verify rule	Rule displayed
curl http://<server-ip>:8080 (from another host)	Test connectivity after blocking	Connection failed
sudo firewall-cmd --remove-rich-rule=...	Remove temporary block	Rule removed successfully

4. List all currently active firewalld zones and display the services and ports allowed in the public zone.

Ans :

Command	Purpose	Example Result
sudo firewall-cmd --get-active-zones	List active zones	public active on eth0
sudo firewall-cmd --zone=public --list-all	Show complete public zone configuration	Services and ports displayed
sudo firewall-cmd --zone=public --list-services	List allowed services	ssh dhcpv6-client http

Command	Purpose	Example Result
<code>sudo firewall-cmd --zone=public --list-ports</code>	List allowed ports	8080/tcp

5. Remove a custom rule you added earlier (such as the HTTP port 80 rule) and verify that the change has taken effect using firewall-cmd.

Ans :

Command	Purpose	Result
<code>sudo firewall-cmd --permanent --remove-service=http</code>	Remove HTTP service rule	success
<code>sudo firewall-cmd --reload</code>	Apply firewall changes	success
<code>sudo firewall-cmd --list-services</code>	Verify allowed services	http no longer listed
<code>sudo firewall-cmd --zone=public --list-all</code>	Verify zone configuration	HTTP rule removed